

A Zonally-Averaged Model of the Meridional Overturning Circulation

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ABSTRACT: Current conceptual theory of the ocean meridional overturning circulation (MOC) describes an interaction between a re-entrant southern channel and a northern sinking region, coupled through a basin with uniform upwelling through flat isopycnals. Near the surface and bottom of the real ocean basin, sloping isopycnals induce flows that these models are unable to resolve by construction. Here, a model that relaxes the flat assumption made on the basin density is proposed, where the basin overturning is related to the meridional gradient of density through a local thermal wind balance. The simple 2D model is tested against idealized 3D simulations from a general circulation model, and is found to reproduce the zonally averaged dynamics. Leveraging the low numerical cost of the model, the effects of vertical mixing and surface buoyancy forcing on the MOC are explored. Results show that the strength and depth of the mid-depth cell generally increase with vertical mixing, consistent with previous studies. For a fixed vertical mixing profile, the depth of the mid-depth cell primarily decreases with the buoyancy contrast across the entire meridional domain, while its strength primarily increases with the shared range of densities between the basin and channel.

1. Introduction

The global ocean meridional overturning circulation (MOC) is a major component of Earth’s climate, which plays a crucial role in controlling the amount of heat and carbon stored in the deep ocean and exchanged with the atmosphere (e.g., Winton et al. 2013; Frölicher et al. 2015; Marshall et al. 2015; Watson et al. 2015; Liu et al. 2017). The MOC has two interconnected cells; a clockwise mid-depth cell linked to the formation of North Atlantic Deep Water (NADW) and a lower anticlockwise cell associated with Antarctic Bottom Water (AABW) (Lumpkin and Speer 2007; Talley 2013). We will loosely refer to these as the Atlantic component of the MOC (AMOC) and the Abyssal circulation, respectively.

While complex three-dimensional models are useful for quantitative predictions, simpler conceptual models allow us to identify the dominant physical mechanisms and study their effects on the circulation. The conceptual theory of the MOC (and our consequent understanding of these processes) has greatly evolved since the pioneering works of Stommel (1961) and Munk (1966). Historically, zonally-averaged models (e.g., Marotzke et al. 1988; Wright and Stocker 1991; Wright et al. 1995, hereafter denoted MWW) have captured the continuous structure of the MOC by leveraging the geostrophic thermal wind equations, whereby a zonal (meridional) pressure gradient is associated with a meridional (zonal) flow (Vallis 2017). In fact, the thermal wind equations have been used to infer the strength of the AMOC in the real ocean by observing the zonal pressure difference across the Atlantic basin (e.g., McCarthy et al. 2012).

The zonally-averaged thermal wind equations however are not closed since the zonal pressure gradient across the basin is generally unknown. To overcome this, MWW provided various parameterizations to relate the meridional flow to the meridional pressure (or density) gradient. These generally add dissipation of momentum such that the thermal wind equations can be coupled together. Marotzke et al. (1988) used enhanced vertical dissipation of momentum¹ to balance the pressure gradient in the meridional momentum equation. Wright and Stocker (1991) then proposed a linear relationship² between the zonal and meridional pressure gradients, which held in a general circulation model. Wright et al. (1995) instead provided a new closure including non-local effects, where vorticity is dissipated in the western boundary. Related work has also been carried out more recently (e.g., Callies and Marotzke 2012; Sévellec and Fedorov 2016).

While this earlier work focused on the overturning in a basin, where sinking at high latitudes is balanced by diffusive upwelling at low latitudes, it was later recognized that upwelling in the Southern Ocean (SO) plays a key role in the dynamics of the MOC (e.g., Toggweiler and Samuels 1995; Gnanadesikan 1999; Marshall and Radko 2003). The overturning circulation in the SO is considerably different than that of the Atlantic or Indo-Pacific basins due in part to its zonally re-entrant geometry. Geostrophic flows generated by meridional density changes in the SO contribute to the Antarctic Circumpolar Current (ACC) as opposed to the MOC in a closed basin (e.g., Gnanadesikan 1999; Nikurashin and Vallis 2012). Instead, the meridional overturning in the SO is driven by wind and eddies, which combine to produce upwelling of deep water along

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¹See also Sakai and Peltier (1995)

²later explained by Wright et al. (1998)

sloping isopycnals and the formation of northern intermediate waters and AABW (e.g., Webb and Sugimotohara 2001; Marshall and Radko 2003; Marshall and Speer 2012).

The need to couple distinct dynamical regions together gave rise to the current conceptual theory, which describes the MOC as the interaction between: (i) the Antarctic Circumpolar Current (ACC) where the residual between the wind- and eddy-driven transports induces a circulation following sloping isopycnals (Webb and Sugimotohara 2001; Marshall and Radko 2003) (ii) a basin region, where buoyancy is vertically diffused across flat isopycnals³ (Samelson 1999; Nikurashin and Vallis 2011), and (iii) a northern convective region, where horizontal gradients of density generate the MOC through thermal wind balance (e.g., Gnanadesikan 1999; Nikurashin and Vallis 2012; Jansen and Nadeau 2019). Coupling these three regions leads to a predictive theory for the global overturning circulation and stratification (Gnanadesikan 1999; Nikurashin and Vallis 2012, hereafter denoted by GNV).

To couple the northern convective region to the southern channel, GNV assume that the density of the basin in between is independent of latitude. While this is reasonable in the mid-depths, according to observations of ocean density (Koltermann et al. 2011), the assumption breaks down near the bottom of the ocean, where isopycnals incrop into the ocean floor. The flow and stratification in the associated abyssal boundary layer (ABL) cannot be resolved by the GNV models. Similarly, the models cannot represent the outcropping of isopycnals at the surface⁴.

The goal of this paper is to develop a conceptual model of the MOC that relaxes the assumption of flat basin isopycnals to revisit the continuous description of MWW type zonally-averaged models. To do this, we build on the current conceptual theory of GNV, which couples a zonal thermal wind flow to the MOC through western boundary currents only at high northern latitudes. We generalize this idea to all basin latitudes to build a global definition of a thermal-wind flow everywhere outside the channel. We then use uniform upwelling as an additional basin flow component to match the transport in the basin to that of the southern channel at their interface.

The paper is organized as follows. Section 2 describes the model, presenting the theory of the thermal wind closure and the numerical implementation. Section 3 validates the model by comparing predictions from the conceptual model against equivalent idealized simulations using a three-dimensional ocean-only general circulation model (GCM). In Section 4, we use the model to explore the circu-

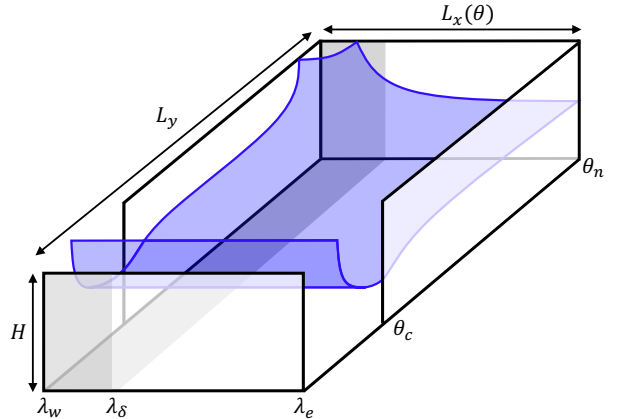


FIG. 1. Idealized model geometry (in spherical coordinates) of a single closed basin, in a shallow spherical sector of the Earth, connected to a re-entrant circumpolar channel at latitude θ_c , both of the same uniform depth H . A typical density surface is shown in blue. The basin region ($\theta_c < \theta < \theta_n$) is here divided into a western boundary layer (shaded gray) and an interior at longitude λ_δ (c.f. Fig. 2).

lation across a wide parameter space in vertical mixing and surface buoyancy forcing. We then conclude in Section 5.

2. Model theory

We here derive the equations for the conceptual 2D model of the global meridional overturning circulation, building in particular on previous work by Nikurashin and Vallis (2012). For simplicity, here we focus on a single basin model of the MOC. Nadeau and Jansen (2020) demonstrated that, to zeroth order, the zonally averaged global overturning circulation is essentially independent of whether the ocean comprises a single basin or multiple interconnected basins. However, a second basin can be added in future work by building on the theory of Nadeau and Jansen (2020).

Consider then a single closed basin on a shallow spherical sector of the Earth, extending from latitude θ_n across the equator and connecting to a zonally re-entrant circumpolar channel of the same uniform depth H at latitude θ_c (see Fig. 1). The western and eastern basin walls are at longitudes λ_w and λ_e , respectively, for a zonal basin width $L_x = a\Delta\lambda\cos\theta$ (where $a = 6378$ km is the Earth's radius, and $\Delta\lambda = \lambda_e - \lambda_w$), which is wider at the equator than at the poles.

We further assume in this work a linear equation of state for density (and therefore buoyancy b) at constant salinity and with thermal expansion coefficient $\alpha_T = 2 \times 10^{-4}$ ($1/^\circ\text{C}$). Since only variations in b are relevant to the dynamics, buoyancy is defined up to a constant which we choose such that buoyancy zero is associated with waters of temperature 0°C . Then, for the scope of this paper, buoyancy and temperature are related by $b = g\alpha_T T$ with $g = 9.81\text{ m/s}^2$.

³See also Shakespeare and Hogg (2012); Thompson et al. (2016).

⁴Wolfe and Cessi (2011) used a 2D model of the basin that considered sloping isopycnals in the basin. In that model the overturning circulation in the basin is described only via eddy- and wind-driven components, with an imposed vertical structure to the wind-driven component which depends on the stratification.

a. Time evolution of buoyancy

Using the Transformed Eulerian Mean framework (Andrews and McIntyre 1976), the zonally averaged buoyancy budget in spherical coordinates (latitude θ , longitude λ , and local vertical z) reads

$$\partial_t \bar{b} + \frac{v^\dagger}{a} \partial_\theta \bar{b} + w^\dagger \partial_z \bar{b} = \partial_z \left(\kappa \partial_z \bar{b} \right) + \text{conv.} \quad (1)$$

for an effective vertical diffusivity $\kappa(z)$, and where the residual velocity $(v^\dagger, w^\dagger)$ advects the zonally-averaged buoyancy $\bar{b}$ in the latitude-depth plane. The effects of convection are represented by the last term (conv.). Accounting for the variable zonal width of the basin, the residual velocity may be expressed in terms of a streamfunction Ψ (in m^3/s) for the zonally integrated flow⁵,

$$(v^\dagger, w^\dagger) = \left(-\frac{1}{L_x} \partial_z \Psi, \frac{1}{a L_x} \partial_\theta \Psi \right), \quad (2)$$

where $L_x = a \Delta \lambda \cos \theta$ is the latitude-dependent zonal width of the basin, and where Ψ is the residual streamfunction resulting from the combined effects of the Eulerian mean $\bar{\Psi}$, and eddy-induced Ψ_{GM} circulation, $\Psi = \bar{\Psi} + \Psi_{GM}$. For the scope of this paper, we neglect the effects of standing eddies which we expect to be especially important near the surface for shallow wind-driven circulation (i.e. gyres). Future work will investigate these fluxes in detail.

The model is posed as an initial value problem, evolving the buoyancy according to (1) and where the residual streamfunction Ψ is diagnosed from $\bar{b}$ in the entire domain, at every time-step. The model is integrated in time until a steady state is reached.

b. Continuous eddy streamfunction across the domain

Eddy buoyancy fluxes are parametrized as in Gent and McWilliams (1990) leading to an eddy-induced circulation opposing the sloping of isopycnals,

$$\Psi_{GM} = -L_x K_{GM} \frac{\frac{1}{a} \partial_\theta \bar{b}}{\partial_z \bar{b}}, \quad (3)$$

with an eddy diffusivity K_{GM} . Allowing sloping basin isopycnals implies the existence of an eddy-induced basin circulation. We then generally have an eddy component to the circulation everywhere in the domain and the dynamical difference between the channel and the basin resides in the mean flow, as we discuss below.

c. Channel dynamics

Since the channel is re-entrant, there is no net zonal pressure difference and so no meridional thermal wind transport. The meridional gradient of buoyancy induces a zonal thermal wind flow which contributes to the ACC rather than the MOC. The mean meridional circulation in the channel is instead driven by equatorward Ekman transport at the surface, (from eastward winds) balanced by a poleward flow at depth to create the so-called Deacon cell (Döös and Webb 1994). This provides the channel mean flow,

$$\bar{\Psi}_{SO} = -L_x \frac{\tau}{\rho_0 f}, \quad (4)$$

for $\tau(\theta)$ the surface wind-stress, $f(\theta) = 2\Omega \sin \theta$ the Coriolis frequency (with $\Omega = 7.2921 \times 10^{-5} \text{ s}^{-1}$), $L_x(\theta)$ the zonal width of the channel, and $\rho_0 = 1035 \text{ kg/m}^3$ the reference density. This mean flow acts to steepen outcropping isopycnals in the channel, and the eddy-induced circulation, Ψ_{GM} – parameterizing eddies arising from baroclinic instability – acts to flatten the isopycnals (Karsten et al. 2002; Marshall and Radko 2003). Our channel dynamics are identical to those described in Nikurashin and Vallis (2011, 2012). The basin dynamics however differ substantially.

d. Thermal wind closure for the basin

We here derive a thermal wind closure for the basin mean flow. For the scope of this section, the velocities discussed are the geostrophic components of the Eulerian mean flow. The governing equations in the basin are thermal wind balance and continuity for a horizontal velocity $\mathbf{u} = (u, v)$ and vertical velocity w , in spherical coordinates at latitude θ , longitude λ , and local vertical coordinate z . They read

$$f \partial_z \mathbf{u} = \mathbf{k} \times \nabla_h b, \quad (5)$$

$$\nabla_h \cdot \mathbf{u} + \partial_z w = 0, \quad (6)$$

where $\mathbf{k}$ is the unit vertical and ∇_h is the horizontal part of the gradient.

We assume that the basin region ($\theta_c \leq \theta \leq \theta_n$) is divided in longitude into a western boundary region ($\lambda_w \leq \lambda \leq \lambda_\delta$) and an interior region ($\lambda_\delta \leq \lambda \leq \lambda_e$) as in Fig. 1 (western boundary shaded in gray). To simplify the dynamics, we assume that *i*) the flow in the western boundary is purely horizontal, *ii*) that the zonal flow out of the western boundary is equal to the zonal mean zonal flow in the basin, and *iii*) that the basin meridional transport (along with any zonal variation of buoyancy) is confined to the western boundary (such that $v = 0$ in the basin interior). We find that these assumptions hold reasonably well in our

⁵similarly to that derived in Callies and Marotzke (2012)

3D simulations, although the neglect of meridional transport in the basin is the most severe limitation and breaks down at high latitudes (see Appendix A).

With these assumptions in mind, continuity (6) in the western boundary, integrated from λ_w up to the interface at λ_δ , provides⁶

$$\frac{1}{a} \partial_\theta \left(\int_{\lambda_w}^{\lambda_\delta} v a \cos \theta d\lambda \right) + u_\delta = 0, \quad (7)$$

using no flow through the western wall ($u|_{\lambda_w} = 0$), $\partial_z w = 0$ in the western boundary (assumption *i*), and defining $u_\delta \equiv u|_{\lambda_\delta}$. Referring to the bottom panels of Fig. 2, this corresponds to balancing meridional changes in the western boundary transport (v) with a zonal flow (u_δ) through the interface at λ_δ , at every depth and latitude. The required vertical transport needed to close the circulation is assumed present in a passive eastern boundary layer.

The horizontal flow in the basin interior is purely zonal and can be found using the zonal average of (5) over the basin width,

$$\partial_z \bar{u} = -\frac{1}{fa} \partial_\theta \bar{b}. \quad (8)$$

To close the problem (in relating the meridional transport to zonally averaged quantities) we now must relate $\bar{u}$ to u_δ , the zonal velocity out of the western boundary region at λ_δ . The results of our GCM simulations indicate that these are approximately equal (assumption *ii*). Recognizing this relationship allows us combine (7) and (8) to write

$$\frac{1}{a} \partial_\theta \partial_z \left(\int_{\lambda_w}^{\lambda_\delta} v a \cos \theta d\lambda \right) = \frac{1}{fa} \partial_\theta \bar{b}. \quad (9)$$

Assuming that most⁷ (assumption *iii*) of the meridional transport occurs in the western boundary (c.f. Wright and Stocker 1991) the meridional overturning streamfunction is given by

$$\partial_z \bar{\Psi}_{TW} = - \int_{\lambda_w}^{\lambda_\delta} v a \cos \theta d\lambda. \quad (10)$$

Then, (9) and (10) provide a third order PDE for the local “thermal-wind” transport in the basin,

$$\frac{1}{a} \partial_{zz} \partial_\theta \bar{\Psi}_{TW} = -\frac{1}{fa} \partial_\theta \bar{b}, \quad (11)$$

⁶Since the domain depth is much smaller compared to Earth’s radius, ($H/a \approx 0.0006$) we approximate the local radius $r = a + H \approx a$

⁷Assuming approximately homogeneous upwelling in the basin, following Stommel and Arons (1959a,b), we could have instead assumed that the western boundary carries twice the overturning transport. Combining this with the consistent assumption that $u_\delta \approx 2\bar{u}$, we would again recover (11).

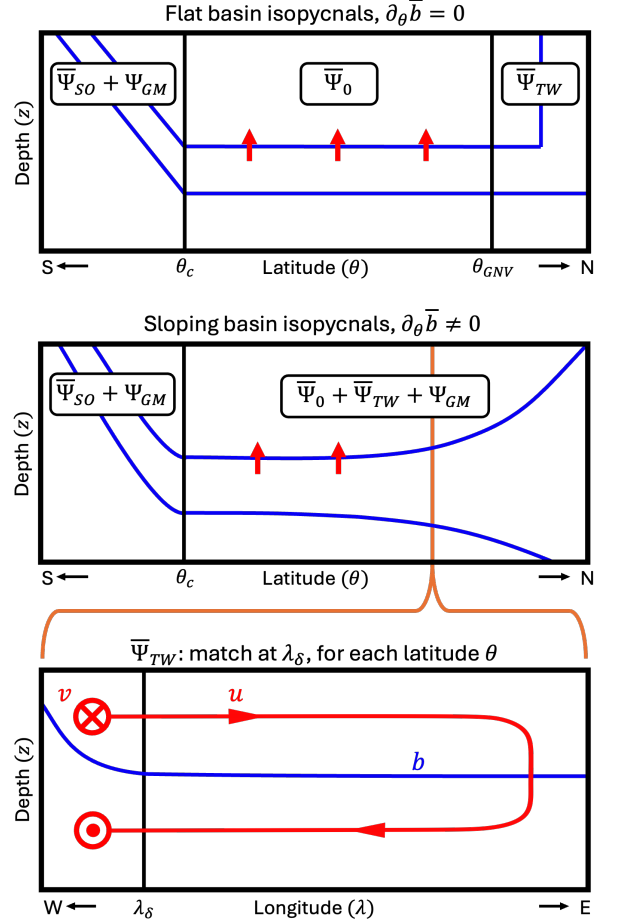


FIG. 2. Schematic comparison between GNV type models (top) and the model discussed herein (middle), where we allow isopycnals (shown as blue lines) to slope in the entire domain outside the channel. GNV only considered a thermal wind contribution north of some high latitude, θ_{GNV} . The bottom panel explains our local thermal wind closure, matching the meridional gradient of the western boundary transport to the zonal flow across the interface, at λ_δ . See Section 2f for the definition of each flow component.

which can be integrated thrice to find a closed form for $\bar{\Psi}_{TW}$:

$$\bar{\Psi}_{TW}(\theta, z) = \int_\theta^{\theta_n} \left(\int_{-H}^z \left(\int_{-H}^{z'} -\frac{1}{fa} \partial_{\theta'} \bar{b} dz'' + C(\theta') \right) dz' \right) ad\theta'. \quad (12)$$

where $C(\theta)$ is an integration constant which will be chosen to satisfy the surface boundary conditions, as we discuss below.

e. Boundary conditions

We require that the circulation Ψ vanishes at all four domain boundaries to ensure no-flow through solid walls.

The eddy component of the circulation Ψ_{GM} is set to zero at all edges to achieve this, as is done in our GCM simulations. The mean-flow in the channel $\bar{\Psi}_{SO}$ is treated similarly at the top and bottom, implying an Ekman layer thickness of one gridpoint. For the basin mean-flow, the chosen integral bounds of (12) impose that $\bar{\Psi}_{TW} = 0$ at the northern edge ($\theta = \theta_n$) and bottom ($z = -H$) of the domain. We chose the integration constant $C(\theta)$ to ensure that the circulation vanishes at the surface ($z = 0$) for every latitude (as in Nikurashin and Vallis 2012). This requires the form

$$C(\theta) = -\frac{1}{H} \int_{-H}^0 \left(\int_{-H}^z -\frac{1}{fa} \partial_{\theta} \bar{b} dz' \right) dz. \quad (13)$$

We further would like to require the circulation to be continuous across the basin-channel interface, but at this stage, we generally do not expect $\bar{\Psi}_{TW}|_{\theta=\theta_c}$ to equal the SO mean flow at θ_c . To ensure flow continuity at the southern end of the basin, we shall follow in the way of GNV by imposing a uniform basin upwelling to mitigate the flow imbalance at the basin-channel interface. We will denote this flow by $\bar{\Psi}_0$. Assuming that the isopycnal slopes are continuous across the interface, the eddy flow component will be as well. Then, for the total flow to be continuous, we must match the basin mean flow $\bar{\Psi}_B \equiv \bar{\Psi}_{TW} + \bar{\Psi}_0$ to the channel mean flow $\bar{\Psi}_{SO}$ at the interface, using $\bar{\Psi}_0$.

To determine the meridional structure of $\bar{\Psi}_0$, we are physically motivated to require that the vertical velocity be uniform in latitude when the basin isopycnals are flat (and so when $\bar{\Psi}_{TW} = 0$ in the whole basin). This ensures that $\bar{\Psi}_0$ will not create sloping isopycnals. Taking into account the meridional change in our basin width, for a uniform upwelling of the zonally averaged buoyancy, we require

$$\frac{1}{a} \partial_{\theta} \bar{\Psi}_0 = w_0(z) a \Delta \lambda \cos \theta \implies \bar{\Psi}_0 = G(z) \sin \theta + H(z), \quad (14)$$

for some unknown functions of depth, G, H . Imposing no flow through the northern wall and continuity of Ψ through the channel interface, we find

$$\bar{\Psi}_0 = \frac{\sin \theta_n - \sin \theta}{\sin \theta_n - \sin \theta_c} (\bar{\Psi}_{SO} - \bar{\Psi}_{TW})|_{\theta=\theta_c}(z), \quad (15)$$

which has a similar form to the uniform basin upwelling considered by GNV. In practice, $\bar{\Psi}_0$ primarily captures the diffusive component of the circulation (see Appendix B).

The advected buoyancy is subject to Dirichlet surface boundary conditions, $\bar{b} = b_s(\theta)$ at $z = 0$, and Neumann boundary conditions at the bottom, $\partial_z \bar{b} = 0$ at $z = -H$, and at the side-walls, $\partial_{\theta} \bar{b} = 0$ at $\theta = \theta_s, \theta_n$.

f. Model summary

To summarize, we evolve the zonally averaged buoyancy in time according to

$$\partial_t \bar{b} - \frac{1}{L_x} \partial_z \Psi \frac{1}{a} \partial_{\theta} \bar{b} + \frac{1}{L_x} \frac{1}{a} \partial_{\theta} \Psi \partial_z \bar{b} = \partial_z (\kappa \partial_z \bar{b}) + \text{conv.} \quad (16)$$

where the overturning streamfunction defined as

$$\Psi = \begin{cases} \bar{\Psi}_{SO} + \Psi_{GM}, & \theta < \theta_c \text{ (channel)} \\ \bar{\Psi}_{TW} + \bar{\Psi}_0 + \Psi_{GM}, & \theta \geq \theta_c \text{ (basin)} \end{cases} \quad (17)$$

is diagnosed from $\bar{b}$ at every timestep, with flow components

$$\bar{\Psi}_{SO} = -L_x \frac{\tau}{\rho_0 f}, \quad (18)$$

$$\bar{\Psi}_{TW} = \int_{\theta}^{\theta_n} \left(\int_{-H}^z \left(\int_{-H}^{z'} -\frac{1}{fa} \partial_{\theta'} \bar{b} dz'' + C(\theta') \right) dz' \right) ad\theta', \quad (19)$$

$$\bar{\Psi}_0 = \frac{\sin \theta_n - \sin \theta}{\sin \theta_n - \sin \theta_c} (\bar{\Psi}_{SO} - \bar{\Psi}_{TW})|_{\theta=\theta_c}(z), \quad (20)$$

$$\Psi_{GM} = -L_x K_{GM} \frac{\frac{1}{a} \partial_{\theta} \bar{b}}{\partial_z \bar{b}}. \quad (21)$$

For comparison to previous studies that have not considered spherical geometry, the cartesian form of the model (in the limit of a constant L_x) is included in Appendix C.

g. Numerics

The numerical scheme integrates (1) in time using a simple first order upwind scheme on a staggered C-grid. The eddy streamfunction Ψ_{GM} is calculated using the average slope of the 4 neighboring grid points in the latitude-depth plane. To tackle the singularity implied by thermal wind balance at the equator, we replace $1/f$ with $f/(f^2 + \varepsilon^2)$, where $\varepsilon = 10^{-6} \text{ s}^{-1}$ is an equatorial friction coefficient⁸, as is used for example in Sévellec and Fedorov (2011). The meridional transport is then continuous and unchanged through the point $f = 0$.

The convection scheme is a modified version of that described in Akmaev (1991) for the ocean, which conserves heat and adjusts the stratification to a prescribed minimum of $\partial_z \bar{b} \approx 2 \times 10^{-8} \text{ s}^{-2}$, approximating the minimum zonal mean stratification diagnosed from the considered GCM simulations. This minimum stratification prevents instabilities generated by the eddy component of the flow acting on neutral layers created by the convection scheme (in which $\partial_z \bar{b}$ would otherwise be 0).

⁸Our results are not sensitive to the specific choice of ε , as long as $\varepsilon \ll f$ over most of the domain and large enough to ensure numerical stability.

3. Validation with a 3D GCM

To validate the 2D model, we directly compare its solutions with equivalent 3D ocean-only simulations using the MIT general circulation model (MITgcm, Marshall et al. 1997).

a. Setup

The domain is idealized as in Fig. 1 spanning $\theta_s = -68^\circ$ to $\theta_n = 68^\circ$ in latitude, and $\Delta\lambda = 60^\circ$ in longitude. The boundary between the open channel and enclosed basin is at $\theta_c = -46^\circ$ latitude and the entire domain has a flat bottom at depth $H = 4$ km. The numerical grid has a relatively coarse 2° horizontal resolution and 40 levels in the vertical, with variable thickness from 37 m at the surface to 159 m at the bottom.

Baroclinic eddies are parameterized with the advective form of the Gent and McWilliams (1990) scheme, with constant eddy diffusivity $K_{GM} = 10^3 \text{ m}^2/\text{s}$ and slope clipping as in Cox (1987) for stability. Temperature advection is done with second order flux limiters and convection is parameterized with locally enhanced vertical diffusion. Note that the convection and advection schemes in MITgcm are different than those in the 2D model described above. However, baroclinic eddies are handled in the same way and with the same parameters. The corresponding grid of the 2D model has the same domain and resolution.

To test the capabilities of the 2D model described above, we vary the vertical diffusivity profile in (1), according to

$$\kappa(z) = \kappa_b \exp\left(-\frac{z+H}{l_\kappa}\right) \quad (22)$$

in both the bottom value $\kappa_b = \kappa(z = -H)$ and in the e -folding scale l_κ . Note that the exponential profile of $\kappa(z)$ is representative of the basin-averaged vertical diffusivity, where the scale l_κ is expected to capture bottom-enhanced mixing due to rough topography (e.g., Nikurashin and Ferrari 2010b,a; Waterhouse et al. 2014; Rogers et al. 2023). A (zonally uniform) sea-surface-temperature (SST) restoring profile is applied in the 3D simulations with a very small restoring timescale of 1 day to simulate the Dirichlet surface boundary conditions used in the 2D model. We vary the SST profile $T(\theta)$ in its northern endpoint $T_N \in [0.1^\circ\text{C}, 2^\circ\text{C}]$ and equator $T_{eq} \in [10^\circ\text{C}, 50^\circ\text{C}]$ values, holding the southern value fixed $T_S = 0^\circ\text{C}$ (top panel of Fig. 3). A wind-stress profile is applied only over the channel, vanishing at the channel-basin interface, and with peak strength $\tau_0 = 0.1 \text{ N/m}^2$ (bottom panel of Fig. 3).

All simulations are initially set to a uniform temperature of 10°C and let to equilibrate under the SST and wind forcing profiles described above for at least 10 000 years, after which the solution is stationary.

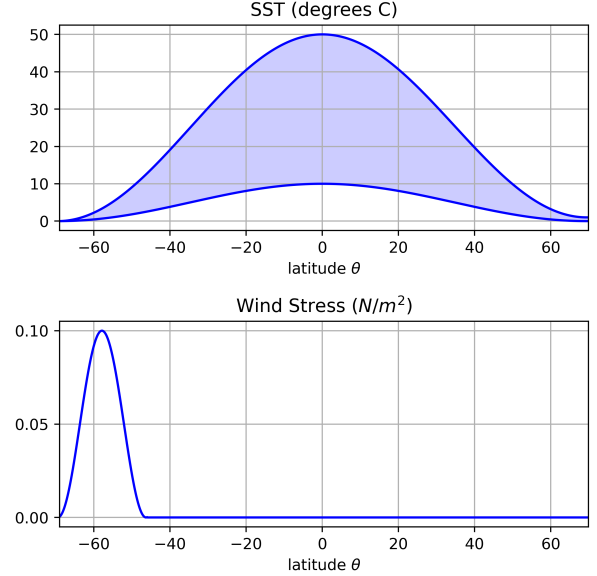


FIG. 3. Surface forcing considered in the simulations herein: (top) SST profile range $T(\theta)$, which has a $\cos^2 \theta$ shape, with $T_{eq} \in [10^\circ\text{C}, 50^\circ\text{C}]$ and $T_N \in [0.1^\circ\text{C}, 2^\circ\text{C}]$, (bottom) wind stress $\tau(\theta)$, nonzero only over the channel region, south of 46°S .

b. 2D vs 3D Comparison

Figure 4 compares nine cases of the 3D MITgcm simulations (top) with the analogous 2D simulation from the conceptual model (bottom), subject to the same surface forcing and diffusivity profiles. Varying l_κ and $\Delta T_{NS} = T_N - T_S$, we find that the 2D model can reproduce the zonally-integrated 3D circulation and zonally-averaged stratification, and their response to changes in the northern SST and e -folding scale of the diffusivity.

Notice that the structure of the ABL and the incropping of isopycnals into the ocean floor is well-resolved in the 2D model. The same is true near the basin surface, where isopycnals may outcrop over the entire length of the domain. In some of our simulations, the abyssal circulation has a local maximum (in the absolute value) in the basin. The associated downwelling in the southern part of the basin is due to a bunching of incropping isopycnals near the bottom of the basin-channel interface, which leads to a bottom-intensified stratification and hence downwelling. This local maximum in the basin can also be seen in other 3D simulations (e.g., Jansen and Nadeau 2016; Jones and Abernathey 2019; Chang and Jansen 2021) and ocean state estimates (e.g., Cessi 2019), and hence does not appear to be an inherently unrealistic result.

Some notable differences nevertheless exist between the 2D and 3D results. As compared to the GCM solutions, the 2D model has: *i*) a slightly weaker channel circulation, *ii*) the maximum depth of the mid-depth cell at slightly lower

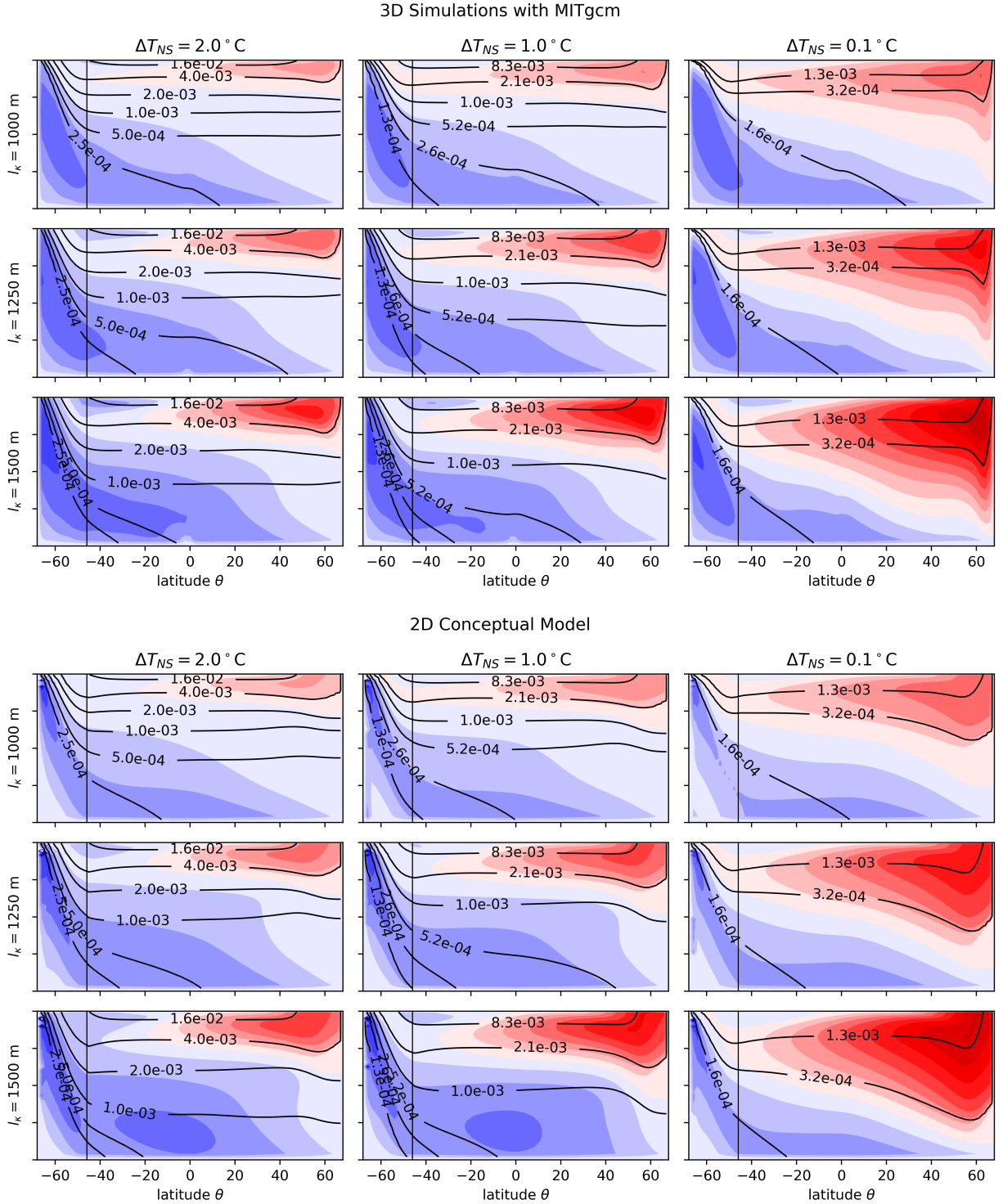


FIG. 4. Comparison between (top) 3D simulations in MITgcm and (bottom) the 2D model for varying northern surface temperature T_N , and diffusivity profiles with varying decay scale, l_k , while keeping $T_{eq} = 30^\circ \text{C}$, $T_S = 0^\circ \text{C}$ and $\kappa_b = 2 \times 10^{-4} \text{ m}^2/\text{s}$ fixed in all simulations. The filled contours indicate the total overturning streamfunction Ψ (calculated in depth space) with contour intervals of 1 Sv, red denoting a positive (clockwise) circulation. Black lines depict selected isopycnals, labeled with their associated buoyancy (in m/s^2).

latitudes, and *iii*) a slightly warmer deep ocean. We believe these differences are due to zonal asymmetries in the 3D density where convection occurs only at select longitudes, whereas the 2D model convects the entire associated latitude circle. The choice of convection scheme may also play a role, leaving opportunities to refine the model in future work to address these differences.

The 2D model is also able to accurately reproduce the vertical structure of the 3D stratification and overturning. Figure 5 shows a basin average (between -46° and 46° latitude) of the stratification $N^2 = \partial_z \bar{b}$ and the overturning Ψ for the same simulations depicted in Fig. 4. Line colors indicate the value of $T_N = \Delta T_{NS}$ (since $T_S = 0^\circ\text{C}$) and columns correspond to the different values of l_k . Comparing the vertical structure of the 2D model (solid lines) with their MITgcm analogue (dashed) we confirm that we can reproduce the vertical structure of Ψ and N^2 , in spite of the differences mentioned above.

Notice that the overturning streamfunction Ψ (in the bottom panels of Fig. 5) changes rapidly at the bottom of the ocean over the last gridpoint. The flow component responsible for this apparent discontinuity is the eddy-component Ψ_{GM} , which is rather strong near the bottom in all considered simulations, but goes to zero at solid boundaries in both the 2D model and the GCM.

Overall, the response of the 2D model to different surface forcing and diffusivity profiles accurately follows that of the MITgcm. We further confirmed that it agrees with 3D simulations at different values of T_{eq} , l_k , and other strengths of the channel wind-stress τ and K_{GM} (not shown). The model resolves the incropping (in the ABL) and outcropping (thermocline) of isopycnals over the entire domain and therefore adds realism over the current state of zonally averaged models that have assumed flat basin isopycnals.

The model is very inexpensive to run (on the order of 100 times faster than the equivalent MITgcm configuration described above for the same time-step). Note that increasing the basin width (L_x) towards global scale 3D simulations, while retaining the same horizontal resolution, becomes very expensive in GCMs. In the 2D model described above, L_x instead does not significantly affect the compute time of a simulation (only through numerical stability considerations).

4. Sensitivity to vertical mixing and surface buoyancy

Since the 2D model is numerically inexpensive, we can use it to quickly and easily test the MOC's dependence on a wide range of parameters. We here focus on the diffusivity profile (by varying the e -folding scale l_k and bottom value κ_b) and on the surface buoyancy profile (comparing the effects of changes in different temperature contrasts).

To measure the system's response to these changes, we will focus on the following diagnostics: $\langle \Psi_{max} \rangle$ and

$\langle h_{max} \rangle$ are the strength of the mid-depth cell (i.e the maximum of $\Psi(z)$) and the depth of the interface between the mid-depth and lower cells (where $\Psi(z) = 0$), both averaged in latitude between 40°N and 50°N . $-\langle \Psi_{min} \rangle$ is the strength of the lower cell, taken as the maximum in depth of $-\Psi$, averaged in latitude over the southern part of the basin, between the channel interface and the equator, and $\langle N^2 \rangle$ is the abyssal stratification, averaged in latitude from -46° to 46° , and under 2 km depth to the ocean floor.

a. Vertical Mixing

Vertical mixing has been argued to be one of the main drivers of the MOC (e.g., Gnanadesikan 1999; Ito and Marshall 2008; Nikurashin and Vallis 2011, 2012; Mashayek et al. 2015). To test the influence of vertical mixing on the MOC, we vary the e -folding scale l_k and bottom value κ_b of $\kappa(z)$, while holding the surface buoyancy forcing fixed.

1) MID-DEPTH CELL

We find that the AMOC strengthens and deepens with both l_k and κ_b in our simulations (Fig. 6 a,b) both of which generally increase the bulk amplitude of vertical mixing. Colored circles in each panel confirm that the equivalent quantities calculated from the GCM agree well with the 2D simulations. The strengthening and deepening of the AMOC in response to enhanced diapycnal mixing is consistent with GNV, who argued that the AMOC has a diffusive component which is expected to increase in strength and depth with vertical mixing (although they consider a constant vertical diffusivity).

Despite the vertical structure in $\kappa(z)$, we find that $\langle h_{max} \rangle$ and $\langle \Psi_{max} \rangle$ scale well with the value of κ at around $z = 1$ km depth, an approximation of the diffusivity across the pycnocline. This agreement can be seen in how the black contours of $\kappa|_{z=1\text{km}}$ follow those of $\langle h_{max} \rangle$ and $\langle \Psi_{max} \rangle$. Though GNV derive scaling laws for the strength and depth of the AMOC in terms of a constant κ , a simple power-law scaling is only found in the limit of no wind-driven upwelling in the SO, which is not applicable here; our basin circulation is not always much stronger than that of the channel. The consequent interaction between the adiabatic and diffusive processes therefore complicates the quantitative relationship between the diffusivity and AMOC strength. Consistently, we do not find a clear power-law relationship between the overturning and the diffusivity here (not shown),

2) LOWER CELL

Our simulations show that the strength and stratification of the lower cell both increase with κ_b and l_k (Fig. 6 c,d). As both κ_b and l_k generally increase the bulk amplitude of vertical mixing, our results generally agree with previous studies that considered the abyssal cell in isolation, with

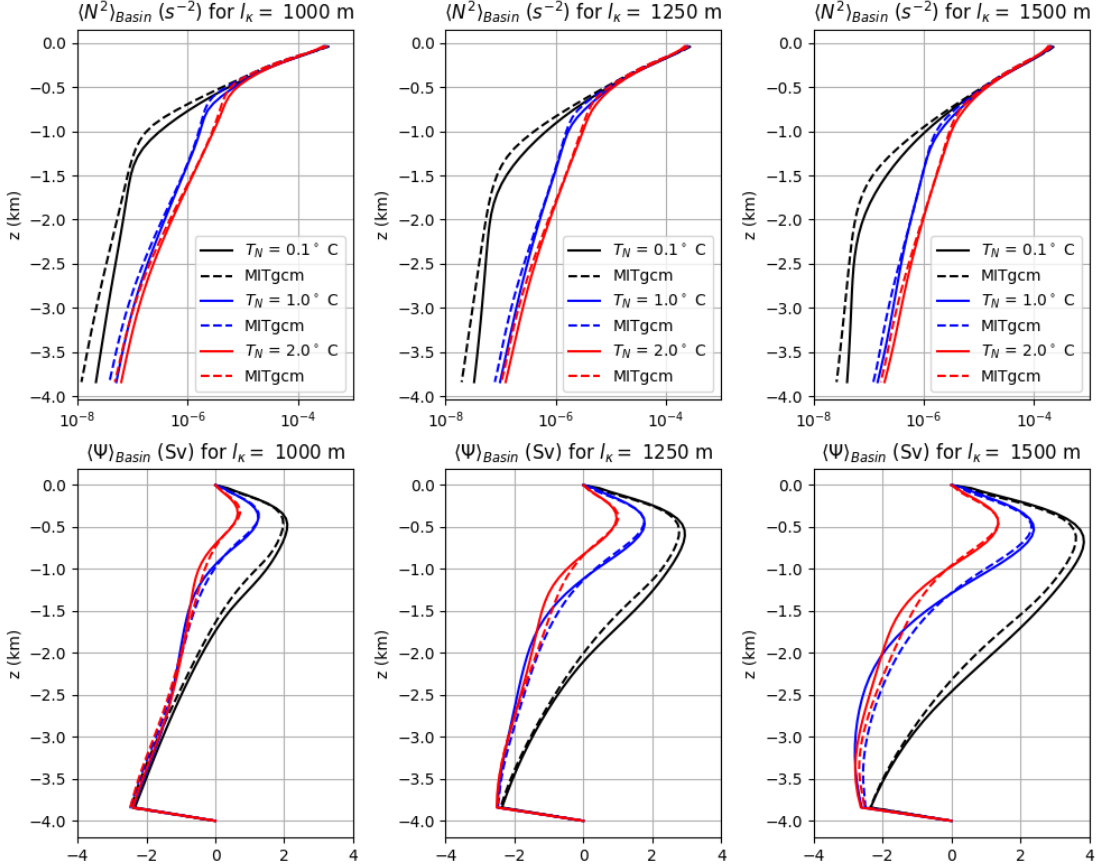


FIG. 5. Vertical profiles for the basin-averaged (from 46°S to 46°N) Brunt-Vaisala Frequency $N^2 = \partial_z b$ (top) and overturning streamfunction Ψ (bottom), from the simulations depicted in Fig. 4 above. Columns show varying l_k and the colors in each column show results with different T_N (see legend). The 2D model (solid line) agrees well with the corresponding MITgcm run (dashed) in all considered parameter regimes.

constant κ (e.g., Ito and Marshall 2008; Nikurashin and Vallis 2011). These theories generally predict that the strength of the lower cell depends on vertical mixing as $\Psi_{min} \propto -\kappa^{1/2}$ when $|\Psi| \ll |\Psi_{GM}|$ or as $\Psi_{min} \propto -\kappa$ when $|\Psi| \approx |\Psi_{GM}|$ (see Chang and Jansen 2021).

Since we have vertical structure in $\kappa(z)$, we find that the lower cell strength (Fig. 6c) scales well with κ at $z = 2$ km (notice the black contours of $\kappa|_{z=2km}$ broadly aligning with the streamfunction contours underneath). Indeed, we find an approximate $-\langle \Psi_{min} \rangle \propto \sqrt{\kappa|_{2km}}$ relation (not shown) in general agreement with previous studies. This relation is less true however as we approach smaller l_k and larger κ_b where naturally κ_b becomes more important near the bottom of the ocean.

Mashayek et al. (2015) considered vertical variation in $\kappa(z)$ in their described structure of the lower cell⁹. They propose three stacked layers: the ABL in which isopycnals incrop into the ocean floor, a deep layer (DL) with enhanced mixing above the ABL, and an adiabatic layer at

the top. The lower cell in this model is contained within both the ABL and the DL, where its strength is expected to scale with $\kappa_b^{1/2}$ in the ABL and l_k in the DL. While Mashayek et al. (2015) consider the strength of the lower cell separately in the DL and the ABL, we find that when l_k is small (and so when the DL flow is weak), then κ_b predominately controls the strength of the lower cell.

We further find that the deep ocean basin stratification $\langle N^2 \rangle$ generally increases with diffusivity in both κ_b and l_k . While the aforementioned studies typically do not include a scaling prediction for the stratification amplitude, they do provide a prediction for the depth scale of the stratification. In fact, Mashayek et al. (2015) predicts that this depth scale, diagnosed as $h = \partial_z \bar{b} / \partial_{zz} \bar{b}$, should scale with $\kappa_b^{1/2}$ in the ABL and l_k in the deep layer (the same functional dependence on these parameters as the strength of the lower cell in both regions). While the deep stratification does increase in amplitude with both parameters, we do not find such clear power law relationships for $\langle N^2 \rangle$ in our simulations.

⁹See also Baker et al. (2021)

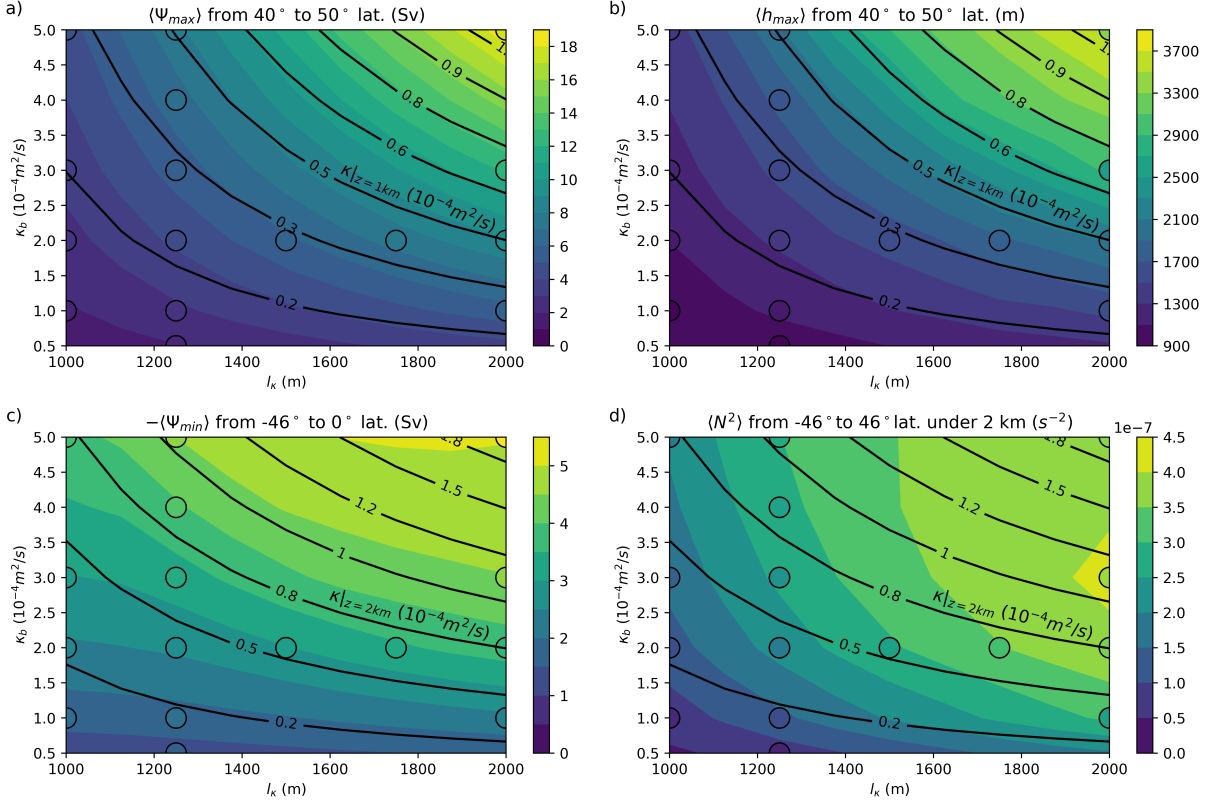


FIG. 6. Filled contours of (a) AMOC strength, (b) AMOC depth, (c) lower cell strength, and (d) lower cell stratification, for various diffusivity profiles $\kappa(z)$ as a function of I_k and κ_b . Diagnostics are calculated from multiple runs of the 2D model with $T_S = 0^\circ\text{C}$, $T_{eq} = 30^\circ\text{C}$, and $T_N = 1^\circ\text{C}$. Colored circles indicate the associated value (calculated in the same way) from the analogous 3D MITgcm simulation. In the top row, black contours indicate the value of $\kappa(z)$ at 1 km, which is a good predictor of the depth and strength of the mid-depth cell. In the bottom row, black contours of κ are given at 2 km, which is a good predicting variable for the strength of the lower cell.

b. Surface forcing

In addition to vertical mixing, the surface buoyancy forcing is expected to affect the global properties of the MOC (e.g., Kamenkovich and Goodman 2000; Wolfe and Cessi 2011; Nikurashin and Vallis 2011; Mashayek et al. 2015; Jansen and Nadeau 2016; Baker et al. 2020; Sun et al. 2020; Chang and Jansen 2021).

The surface buoyancy contrasts of interest in these studies include: the range of surface buoyancy values shared between the channel and northern hemisphere (e.g., Wolfe and Cessi 2011), the buoyancy contrast across the entire meridional domain (e.g., Sun et al. 2020; Nadeau and Jansen 2020), and that across the channel width (e.g., Nikurashin and Vallis 2011; Mashayek et al. 2015; Chang and Jansen 2021). To measure the influence of each of these contrasts, we vary the northern T_N and equatorial T_{eq} surface temperatures in a set of simulations with fixed vertical mixing $\kappa(z)$ and $T_S = 0^\circ\text{C}$. Recall that with constant salinity, isopycnals and isotherms are equivalent. Varying T_{eq} directly varies the surface temperature at the channel

interface T_C (cf. Fig. 3) and so the buoyancy contrast across the channel $\Delta T_{SC} = T_C - T_S$. Varying T_N controls the surface temperature contrasts across the entire domain $\Delta T_{NS} = T_N - T_S$, while both T_N and T_{eq} affect the shared buoyancy range between the basin and channel $\Delta T_{NC} = T_C - T_N$.

1) MID-DEPTH CELL

In our considered parameter range, it is clear that the strength and depth of the AMOC (Fig. 7 a,b) do not depend on only one of these temperature contrasts. Qualitatively, we find that the AMOC generally strengthens and deepens with increasing ΔT_{NC} (the shared range of densities) but shoals and weakens with increasing ΔT_{NS} , in general agreement with previous studies (e.g., Wolfe and Cessi 2011; Nadeau and Jansen 2020; Sun et al. 2020). Colored circles again show that the equivalent quantities from the GCM agree well with the 2D simulations.

We find that the strength of the AMOC increases most sensitively to ΔT_{NC} . This is consistent with Wolfe and

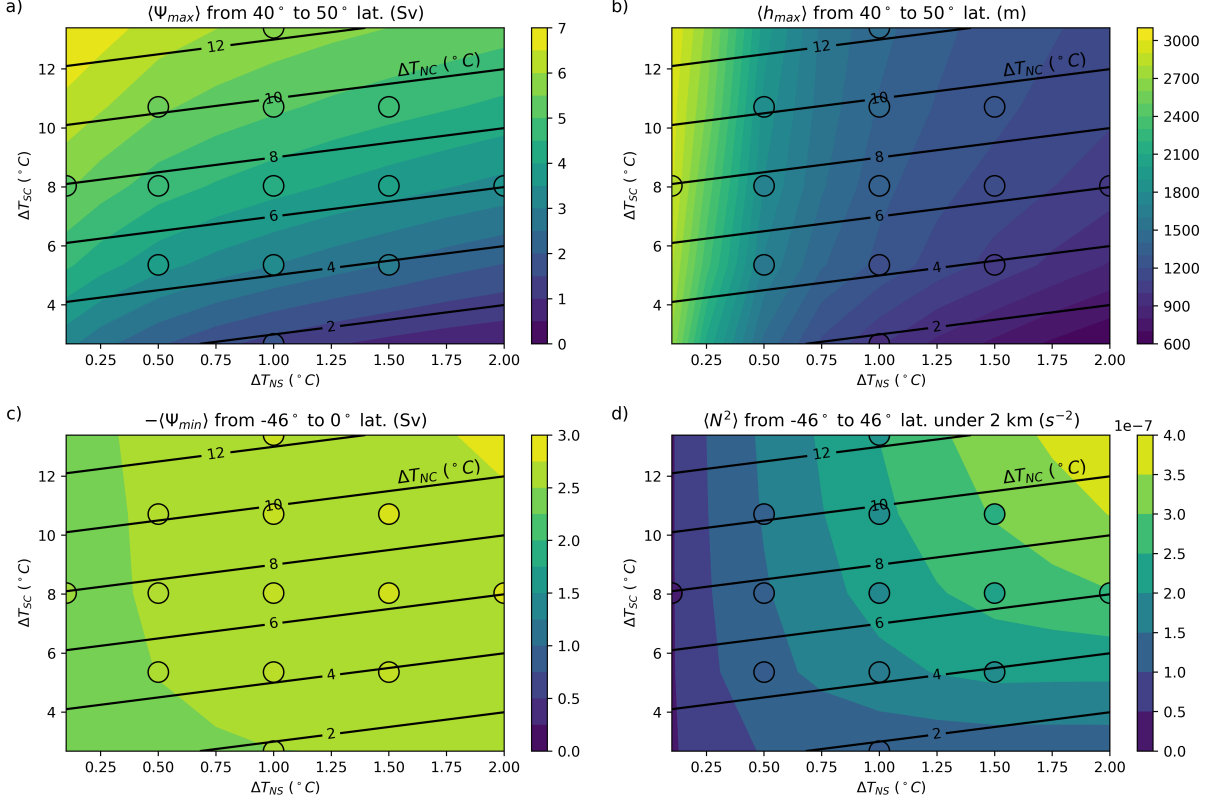


FIG. 7. Same diagnostics as in Fig. 6, but as a function of ΔT_{NS} and ΔT_{SC} , having varied $T_N \in [0.1, 2]$ and $T_{eq} \in [10, 50]$ respectively. Black contours indicate ΔT_{NC} . Colored circles indicate the equivalent quantities from the 3D MITgcm simulation. Diffusivity profiles in all simulations considered here use $\kappa_b = 2 \times 10^{-4} \text{ m}^2/\text{s}$ and $l_k = 1250 \text{ m}$.

Cessi (2011), who argued that the strength of the adiabatic component of the AMOC increases with the shared range of densities which outcrop in both the North Atlantic and the Southern Ocean (ΔT_{NC}).

The depth of the AMOC is instead most sensitive to ΔT_{NS} , which is consistent with Nadeau and Jansen (2020) and Sun et al. (2020). Specifically, the AMOC shoals as northern waters get lighter (warmer in our framework) compared to those at the southern edge of the channel.

Notice that increased surface buoyancy loss due to ice-growth around Antarctica has also been argued to shoal the AMOC (e.g., Jansen and Nadeau 2016; Nadeau et al. 2019). By increasing the surface buoyancy loss in the channel, these studies decrease the minimum buoyancy at the southern end of the channel surface. This increases both ΔT_{SC} and ΔT_{NS} , while keeping the shared range ΔT_{NC} relatively constant. To compare our results to those of Jansen and Nadeau (2016) and Nadeau et al. (2019), we must therefore interpret our results along constant lines of ΔT_{NC} ; and indeed, we find that the AMOC shoals (and weakens) with increasing ΔT_{SC} and ΔT_{NS} along contours

of constant ΔT_{NC} (following the black contours in Fig. 7b).

2) LOWER CELL

We find that the lower cell strength and stratification both increase with ΔT_{SC} and ΔT_{NS} (cf. Fig. 7 c,d), though the lower cell strength varies by only $\sim 0.5 \text{ Sv}$ ($\sim 20\%$) in the considered parameter space.

The increase in the abyssal cell strength with increasing buoyancy contrasts is qualitatively consistent with Kamenkovich and Goodman (2000), Jansen and Nadeau (2016), Baker et al. (2020), and Chang and Jansen (2021), although Nikurashin and Vallis (2011) predict no sensitivity to the buoyancy gradient. Chang and Jansen (2021) showed that the apparent discrepancy between these theories is due, in part, to the fact that the strength of the lower-cell in the channel and in the basin scale differently with the system parameters. The strength of the lower cell in the basin and the deep basin stratification are then expected to increase with integrated surface buoyancy loss, or, equivalently, with increased buoyancy contrast across the abyssal cell (Jansen and Nadeau 2016; Chang and Jansen 2021).

However, it is not entirely clear in how far the buoyancy contrast across the abyssal cell scales with ΔT_{SC} or ΔT_{NS} . The minimum northern temperature essentially provides the temperature at the bottom of the mid-depth cell. If the mid-depth cell is sufficiently deep, this is the same temperature as the top of the abyssal cell. In this case, ΔT_{NS} may be a good predictor of the temperature contrast across the abyssal cell. Conversely, when the AMOC is shallow, ΔT_{SC} may predominately determine this buoyancy contrast. This is qualitatively confirmed by our results, although the sensitivity of the lower cell strength to the surface temperature gradients is found to be relatively small. When the AMOC is deep in Fig. 7b, the contours of the lower cell strength in Fig. 7c become vertical and so are predominately controlled by ΔT_{NS} . Similarly, when the AMOC is shallow, contours of $-\langle \Psi_{min} \rangle$ become horizontal and thus depend mainly on ΔT_{SC} .

The deep stratification similarly depends on both the inter-hemispheric and channel surface buoyancy contrasts in our simulations (Fig. 7d). In particular, we observe that when the AMOC is shallow, increasing the channel surface ΔT_{SC} increases the deep stratification, in qualitative agreement with Chang and Jansen (2021), who considered the abyssal cell in isolation. Conversely, when the AMOC is deep, the domain-wide temperature contrast ΔT_{NS} dominates the deep stratification, consistent with Jansen and Nadeau (2016), who considered the interaction between both the mid-depth and lower cells.

5. Conclusion

We have constructed in this work a continuous zonally-averaged model of the MOC which accounts for meridional variations in basin density using a series of physically-motivated approximations. We have assumed that most of the meridional transport in the basin is concentrated in the western boundary and is coupled to a zonal thermal wind flow in a basin interior to the east. While GNV considered this coupling only at high northern latitudes, we extended the framework to the entire basin region, allowing a thermal-wind flow component at all latitudes (outside the channel). To handle the mismatch between the basin thermal-wind flow and the wind-driven circulation of the channel, we utilized the uniform diffusive upwelling ($\bar{\Psi}_0$) of the GNV models as an additional basin-mean-flow component. The model reproduces the zonally-averaged dynamics of the equivalent 3D GCM simulations with remarkable accuracy. The model also reproduces the 3D model's circulation dependence on considered system parameters (surface buoyancy forcing and vertical mixing) at much lower numerical cost.

There are however some caveats to this study; While we have qualitatively linked $\bar{\Psi}_0$ to the diffusive component of the basin mean flow, as it vanishes in the adiabatic limit of our solutions, the dynamical interpretation and

justification for this flow component remains not fully understood. Further, the model does not currently account for the interaction between the overturning circulation and the wind-driven gyres in the basin. Finally, we have not yet considered the effects of varying salinity (and sea-ice), multiple basins, and the transient response of the model to changes in surface forcing. The model therefore provides a foundation for future work to address these caveats to capture in a simple 2D framework: the structure of the ventilated thermocline (Luyten et al. 1983) and its interaction with the deep ocean circulation, the influence of the salt-advection feedback (e.g., Wolfe and Cessi 2014; Cessi 2018), the various modes and pathways of the circulation (e.g., Ferrari et al. 2017; Nadeau and Jansen 2020), and the transient response of the MOC to surface warming in the context of climate change (e.g., Jansen et al. 2018).

In summary, we have created an efficient tool for the broader scientific community to quickly and efficiently test hypotheses on the dominant physical mechanisms of the MOC without the numerical cost of three-dimensional simulations. The model may also easily be coupled to a similarly idealized atmosphere (with arbitrary meridional structure of the buoyancy at the air-sea interface) to study exchanges between the ocean and the atmosphere in a simple 2D climate model.

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Data availability statement. The code developed in this study is freely available on GitHub at <https://github.com/jonathan-tessier/2D-MOC>. The MITgcm is available at <https://github.com/MITgcm>. Specific configuration files for the simulations are available upon request from JT.

APPENDIX A

Model assumptions

The model assumes that *i*) the flow in the western boundary is purely horizontal, *ii*) that the zonal flow out of the western boundary is equal to the zonal mean zonal flow in the basin, and *iii*) that the basin meridional transport (along with any zonal variation of buoyancy) is confined to the western boundary (such that $v = 0$ in the basin interior). In order to quantify the validity of these assumptions, we have diagnosed the relevant quantities from a typical 3D simulation in Fig. A1.

Assumption *i*) essentially rests in how well (7) holds, where continuity with no vertical transport in the western boundary gives: $-\frac{1}{a}\partial_\theta \int_{\lambda_w}^{\lambda_\delta} v a \cos \theta d\lambda = u_\delta$. We diagnose u_δ as the Eulerian zonal velocity at $\lambda = 10^\circ$ longitude, from the observed width of the western boundary current in that simulation and $-\frac{1}{a}\partial_\theta \int_{\lambda_w}^{\lambda_\delta} v a \cos \theta d\lambda$ is computed using the Eulerian meridional flow in the western boundary current. We find that these two quantities are in fact quite similar (see top- and middle-left panels of Fig. A1). This supports our assumption that the upwelling in the western boundary is small enough to be ignored for the purpose of Eq. (7).

Assumption *ii*) considers that the zonal flow at the eastern edge of the western boundary is equal to the zonally averaged zonal flow, which itself is in thermal wind balance with the zonally averaged meridional buoyancy gradient. We further added in the bottom-left panel of Fig. A1 the zonal mean of the Eulerian zonal velocity $\bar{u}_{MIT}$ to be compared with u_δ in the panel above it. In fact, u_δ and $\bar{u}_{MIT}$ are shown to match reasonably well, although u_δ tends to be somewhat larger over much of the ocean. As discussed in footnote 7 the effect of a larger u_δ and a return circulation in the basin tend to partially cancel. We have further confirmed (not shown) that the zonally averaged zonal flow in the basin, $\bar{u}_{MIT}$ is in thermal wind balance as it matches the predicted zonal flow $\bar{u}$ from integrating $\frac{1}{f_a}\partial_\theta \bar{b}_{MIT}$ in depth.

Assumption *iii*) states that the meridional flow is confined to the western boundary. The right column of Fig. A1 shows the Eulerian meridional velocity v_{MIT} at three different latitudes, $\theta = 10^\circ, 30^\circ, 50^\circ$. At low latitudes, we see that our assumption holds quite well, although at high latitudes the meridional flow in the interior is significant. Thus, we acknowledge that this approximation is severe and holds better at low latitudes. We are currently working on a new version of the model that relaxes this assumption to study the wind-driven gyres.

APPENDIX B

Comments on $\bar{\Psi}_0$

In GNV type models, the dynamics in the basin are assumed to be relatively efficient in keeping isopycnals flat. Taking that limit, the thermodynamics tell us that the upwelling has to be spatially uniform, balancing the mismatch of influx from the channel and northern sinking regions. This motivates the meridional structure of $\bar{\Psi}_0$. While thermal wind in these models is implied to generate any necessary transport even with very small isopycnal slopes, we generally find that there are non-negligible slopes in the basin interior, but that they are too small to provide the flow needed to feed the upwelling. This is why we need to introduce an additional upwelling, $\bar{\Psi}_0$.

Some additional intuition can be gained by studying the case of a purely adiabatic circulation (e.g., Wolfe and Cessi 2011), and progressively adding vertical diffusion to study the behavior of $\bar{\Psi}_0$. This is shown in Fig. B1, for three different values of $\kappa_b \in \{0, 5 \times 10^{-5}, 2 \times 10^{-4}\}$ m²/s, with $l_\kappa = 1250$ m and a fixed surface buoyancy profile. The top row shows a meridional slice of $\bar{\Psi}_0$, $\bar{\Psi}_{TW}$ and their sum (the basin mean flow) at a depth $z = 500$ m. The rows below show the complete 2D structure of $\bar{\Psi}_0$, $\bar{\Psi}_{TW} + \Psi_{GM}$ and the total flow $\Psi = \bar{\Psi}_0 + \bar{\Psi}_{TW} + \Psi_{GM}$, respectively.

In the purely adiabatic case (left column, $\kappa = 0$) and at equilibrium, $\bar{\Psi}_0$ is zero everywhere in the basin. Consequently, $\bar{\Psi}_{TW}$ vanishes at the channel interface θ_c , and the sum $\bar{\Psi}_{TW} + \Psi_{GM}$ (second/third row) describes an adiabatic flow (along isopycnals) in the basin interior. As κ_b is increased in the next two columns, we find that $\bar{\Psi}_{TW}$ does not vanish at the interface, and more so with larger $\kappa(z)$ (see top row, from left to right). Thermal wind and eddies alone are then unable to produce the upwelling necessary to match the dynamics of the channel when κ is non-zero. $\bar{\Psi}_0$ then increases to close the circulation at θ_c . The sum of the other flow components $\bar{\Psi}_{TW} + \Psi_{GM}$ however does not in general remain adiabatic, though this appears approximately true closer to the surface. In the general case of a sloping stratification, $\bar{\Psi}_0$ does not purely translate to the diffusive component of the circulation, but still appears to capture most of the diffusive upwelling.

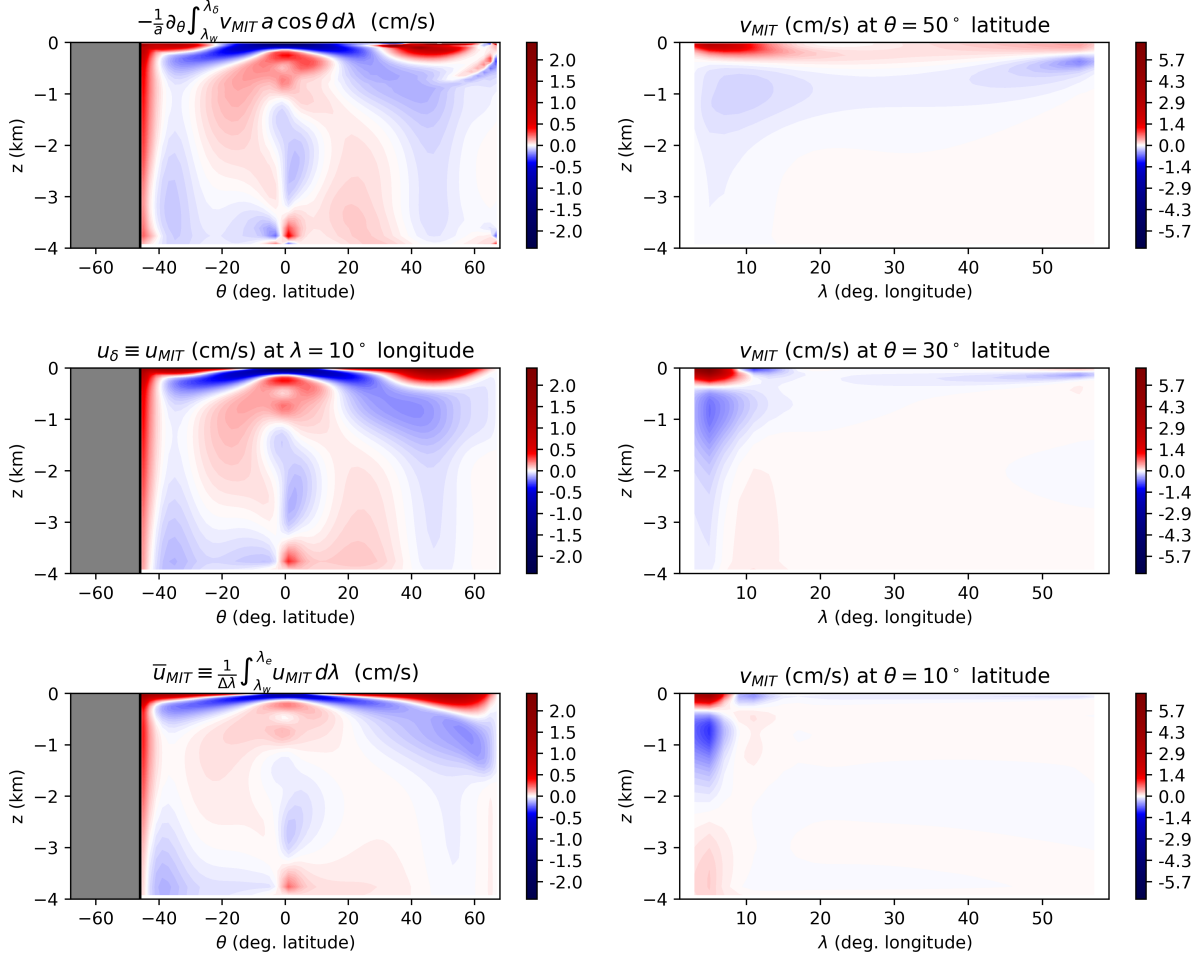


FIG. A1. Various diagnostics calculated from the same GCM simulation as shown in the center panel of Fig. 4, ($T_N = 1^\circ\text{C}$, $T_{eq} = 30^\circ\text{C}$, $l_\kappa = 1250\text{m}$, $\kappa_b = 2 \times 10^{-4} \text{ m}^2/\text{s}$). The top-left and mid-left panels show the two terms from Eq. (7), the bottom-left shows the zonally-averaged zonal flow, and the right column shows the Eulerian meridional velocity at various latitudes.

APPENDIX C

and where the residual overturning streamfunction, still

Cartesian Model

In cartesian coordinates (x, y, z) for a rectangular domain of constant zonal width L_x , the model reduces to evolving the zonally averaged buoyancy in time according to

$$\partial_t \bar{b} + v^\dagger \partial_y \bar{b} + w^\dagger \partial_z \bar{b} = \partial_z (\kappa \partial_z \bar{b}), \quad (\text{C1})$$

where now the zonally *averaged* residual velocity $(v^\dagger, w^\dagger)$ is also divergence free, since the domain has constant width. This allows for the definition of an advective streamfunction $\psi = \frac{1}{L_x} \Psi$ such that

$$(v^\dagger, w^\dagger) = (-\partial_z \psi, \partial_y \psi) \quad (\text{C2})$$

defined as

$$\Psi = \begin{cases} \bar{\Psi}_{SO} + \Psi_{GM}, & y < y_c \text{ (channel)} \\ \bar{\Psi}_{TW} + \bar{\Psi}_0 + \Psi_{GM}, & y \geq y_c \text{ (basin)} \end{cases} \quad (\text{C3})$$

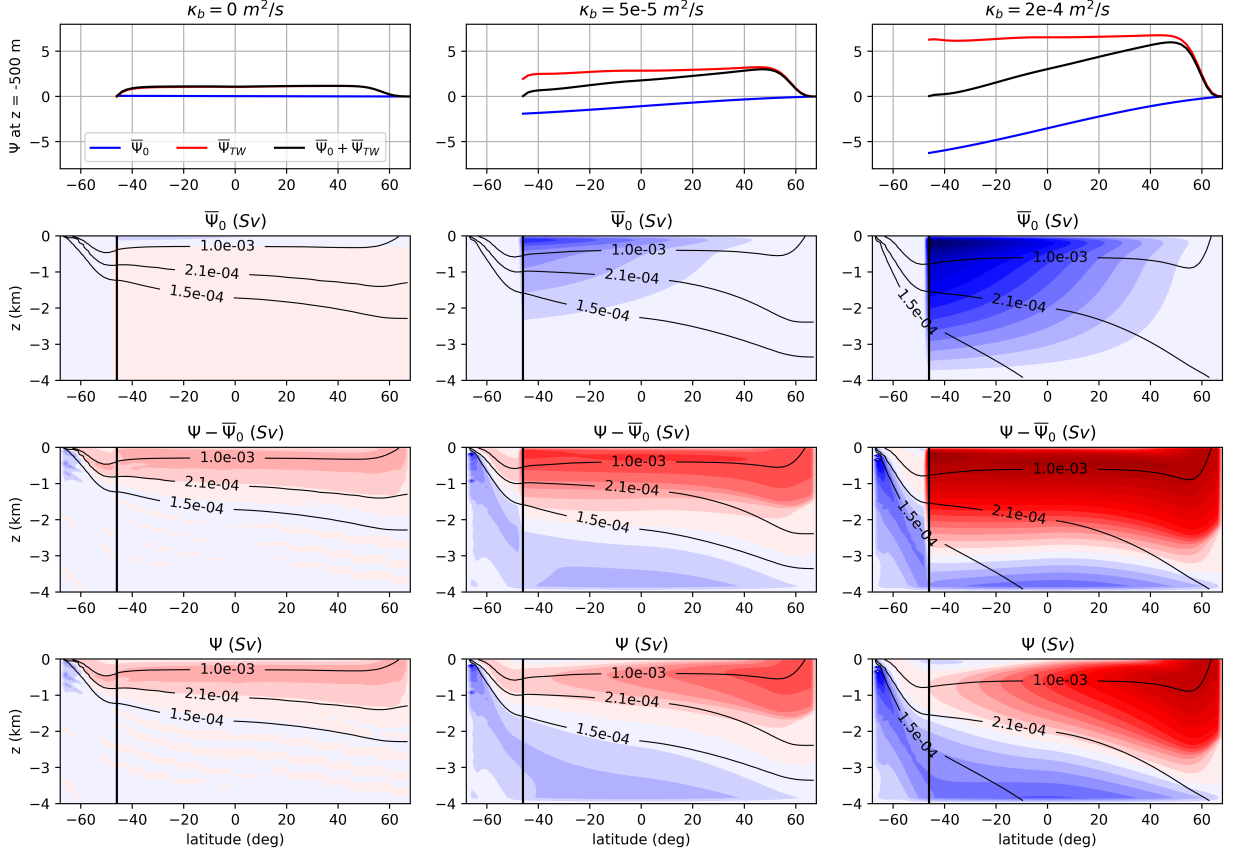


FIG. B1. Three different simulations using the 2D model, for three different values of bottom diffusivity (all with $l_k = 1250\text{m}$): The adiabatic case $\kappa_b = 0$, (left) an intermediate case with $\kappa_b = 5 \times 10^{-5}$ (middle) and a strongly diffusive case with $\kappa_b = 2 \times 10^{-4}$ (right). In all three we used $T_S = 0$, $T_{eq} = 30$, $T_N = 0.05$ deg C. The top row shows a slice of the flow components in each case at depth $z = 500\text{m}$, isolating the $\bar{\Psi}_0$ (blue) and $\bar{\Psi}_{TW}$ (red) components of the basin mean flow $\bar{\Psi}_0 + \bar{\Psi}_{TW}$ (black). The rows beneath show for each case of κ_b the 2D structure of $\bar{\Psi}_0$, $\Psi - \bar{\Psi}_0$ (which equals $\bar{\Psi}_{TW} + \bar{\Psi}_{GM}$ in the basin), and the total overturning Ψ , respectively. In the bottom two rows, the channel flow is the full residual circulation Ψ . Contour intervals are 0.5 Sv, red (blue) indicating positive (negative) values.

is diagnosed from the stratification at every timestep, with flow components

$$\bar{\Psi}_{SO} = -L_x \frac{\tau}{\rho_0 f}, \quad (\text{C4})$$

$$\bar{\Psi}_{TW} = \int_{y_n}^{y_c} \left(\int_{-H}^z \left(\int_{-H}^{z'} -\frac{1}{f} \partial_{y'} \bar{b} dz'' + C(y') \right) dz' \right) dy', \quad (\text{C5})$$

$$\bar{\Psi}_0 = \frac{y_n - y}{y_n - y_c} \left(\bar{\Psi}_{SO} - \bar{\Psi}_{TW} \right) |_{y=y_c}(z), \quad (\text{C6})$$

$$\bar{\Psi}_{GM} = -L_x K_{GM} \frac{\partial_y \bar{b}}{\partial_z \bar{b}}. \quad (\text{C7})$$

This cartesian version may be more easily compared with previous models which neglected meridional variation in the basin width. The flow components considered in Nikurashin and Vallis (2012), as depicted in the top panel

of Fig. 2, are identical to ours in the channel. However, they have restricted the thermal wind component $\bar{\Psi}_{TW}$ to a northern sinking region, extending from some high latitude θ_{GNV} to the northern edge of the domain. Their flat-isopycnal basin then has uniform diffusive upwelling governed by the difference between $\bar{\Psi}_{TW}$ in the northern convective region and $\bar{\Psi}_{SO}$ at the basin-channel interface. In contrast, we have a thermal wind component extending from the northern edge to the channel interface and an eddy-circulation everywhere in the domain. Our $\bar{\Psi}_0$ then only balances the mismatch in the mean flows at the interface and distributes the upwelling everywhere outside the channel.

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